

Aryan Pandey

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EDUCATION

University at Buffalo, State University of New York

Buffalo, NY

Master of Science (M.S.) in Computer Science and Engineering, Research Track

Jan 2025 – Dec 2026 (Expected)

University at Buffalo, State University of New York

Buffalo, NY

Bachelor of Science (B.S.) in Computer Engineering (GPA: 3.7, Dean's List)

Aug 2018 – May 2022

EXPERIENCE

Research Software Engineering Fellow

Jun 2026 – Aug 2026

Princeton University – Princeton Neuroscience Institute

Princeton, NJ

- Enabled researchers to fit cognitive models to behavioral data across multinode HPC (SLURM) clusters instead of a single machine, by building distributed systems infrastructure for parameter estimation in *PsyNeuLink*, an open-source simulation library.
- Designed a Dask-based optimization backend that spins up a worker cluster inside a SLURM allocation, with near-linear throughput scaling to **128+ cores** and reproducible recovery of the same parameters as a serial fit.
- Diagnosed and fixed a latent compilation bottleneck in the library core that recompiled the model on every likelihood evaluation (roughly **8×** per-evaluation overhead), yielding a **5.1×** end-to-end speedup on benchmark fits.
- Extended the backend to hierarchical multi-subject fitting, scaling population-level studies by running each subject's fit as an independent task with models cached per worker to bound memory.
- Verified end-to-end parameter recovery on multi-node runs against an established Bayesian package (HSSM) to confirm statistical correctness, not just throughput.

Software Engineer

Jun 2022 – Apr 2024

Axon

Remote

- Supported development of a large-scale archival system over one of the largest Azure databases in production, using Temporal and Azure Blob Storage to cut cloud storage spend by **\$1.5M/month**.
- Improved evidence-video ingestion throughput by roughly **1.5×**, enabling the service to scale to **250 PB** while holding operational SLOs.
- Held on-call ownership for globally deployed ingestion of body-cam, TASER, and CJIS data, maintaining Dockerized services at **99.99%+** availability.
- Built Scala microservices for back-end infrastructure in a large production codebase, with unit testing, integration testing, and load testing across all services.
- Deployed and released monthly updates across every production region via Kubernetes and Puppet.
- Improved observability across back-end services with Grafana dashboards and restructured Splunk logging, reducing time-to-detect during incidents.

Teaching Assistant – VLSI, Computer Architecture, Computer Security

Aug 2025 – Present

University at Buffalo

Buffalo, NY

- Led labs on CMOS logic and standard-cell design: inverter sizing, NAND/NOR cells, ring oscillators, and SRAM bitcells.
- Taught Cadence Virtuoso schematic, simulation, and layout workflows including parasitic extraction and DRC/LVS sign-off; guided students through timing/power trade-offs, drive-strength selection, and floorplanning for small ASIC blocks.
- Mentored teams through full-flow hierarchical mixed-signal IC projects (DPLLs, TRNGs) from spec to tapeout, reviewing iterations and debugging design and verification issues.
- Authored lab notes, debugging checklists, assignments, and exams; held weekly office hours and ran review sessions across combined undergraduate/graduate sections.

Teaching Assistant – Computer Security, Computer Organization

Aug 2020 – Dec 2021

University at Buffalo

Buffalo, NY

- Ran weekly recitations on CPU architecture, ALU implementation, binary/hex computation, and pipelining; led exam review sessions.
- Covered secure design, malware, risk management, authentication, OS and network security, cryptography, and social engineering; demonstrated common vulnerabilities and their countermeasures.

Software Engineer – Research Internship

Jan 2020 – Aug 2020

University at Buffalo

Buffalo, NY

- Built the C back-end for a portable FASTQ/A DNA analysis machine, including an inotify-based watchdog that detected and routed new files into the analysis pipeline asynchronously.
- Parallelized the pipeline with OpenMP across all cores for an **8×** speedup over the single-threaded baseline.

PROJECTS

Recurrent Differentiable Weightless Networks (RDWN) | *PyTorch, Verilog, Vivado* Feb 2026 – Present

- Added recurrent feedback paths to Differentiable Weightless Networks, a LUT-only architecture that replaces multiply-accumulates with table lookups for orders-of-magnitude cheaper FPGA inference than quantized DNNs; each trained lookup table maps directly onto a physical FPGA LUT-6.
- On Spiking Heidelberg Digits, the recurrent variant reached **82.2%** test accuracy vs. **67%** for the feedforward baseline; **97.75%** on N-MNIST.
- Wrote a PyTorch → Verilog export tool that extracts trained truth tables and learned wiring into synthesizable RTL; demonstrated the model end-to-end on FPGA at **0.6W**.

RL-Based Multi-Objective Neural Architecture Search for SNNs | *PyTorch, REINFORCE* Jan 2026 – Present

- Built an LSTM-controller NAS pipeline (REINFORCE with baseline) that searches spiking network architectures against a joint objective of accuracy, spike sparsity, and temporal coding quality.
- Designed a gated reward that blocks the degenerate solutions where the search converges on networks that never spike, the dominant failure mode for multi-objective SNN search.
- On Spiking Heidelberg Digits the controller converged to a single architecture and outperformed random search under a matched compute budget..

Hardware-Efficient RF Device Fingerprinting | *PyTorch, Norse, NumPy* Nov 2025 – Jan 2026

- Built an end-to-end SNN pipeline from raw complex IQ samples using spatiotemporal receptive fields with LIF/LI dynamics as an alternative to CNN feature extractors for low-power edge inference.
- Matched CNN baselines on 30-device LoRa classification (**99%** validation, roughly **85%** held-out) under the baseline paper's seen/unseen/cross-manufacturer evaluation protocol.
- Trained metric-learning embeddings on the receptive-field features for open-set rogue detection; achieved **0.98 AUC** and **92%** cross-manufacturer generalization.

CUDA Neural Network Optimization for On-the-Fly Training | *CUDA, PyTorch, Triton* Mar 2025 – May 2025

- Extended Hugging Face training to reduce host-device transfers: kept generated samples on the GPU and transferred them only on mismatch, cutting end-to-end training time by roughly **20%**.
- Overlapped H2D transfers, D2H transfers, and computation using pinned memory and multi-stream execution; reduced a stress-test step from **46.71s** to **39.34s**.
- Profiled the pipeline to locate data-movement stalls and GPU-idle regions; improved utilization through stream scheduling and batching changes.

SRAM Physically Unclonable Function | *Cadence Virtuoso, GPDK045* Aug 2024 – Dec 2024

- Designed a 6T-SRAM array exploiting charge-sharing and process variation for PUF-based cryptographic key generation; built transistor-level schematics and layouts in 45nm.
- Characterized bit stability and readout margins via Monte Carlo simulation and post-layout parasitic extraction.

ResearchShelf | *Python, FastAPI, Vue 3, SQLite, Docker* Jul 2026 – Present

- Built and released a self-hosted, multi-user research-paper platform with role-based team libraries, background processing, REST APIs, and a responsive browser-based PDF reader.
- Engineered PDF ingestion across Zotero, DOI/arXiv, Crossref, OpenAlex, and Unpaywall with SHA-256 deduplication and FTS5 full-text search; exposed authenticated workflows to LLM agents via MCP.
- Published to PyPI with automated GitHub releases and pytest/web-build CI covering **29 test modules across 6 OS/Python configurations** (Linux, macOS, Windows; Python 3.11/3.12).

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA, Scala, Java, SQL, JavaScript, Verilog, Bash, ARM/x86/MIPS Assembly

Machine Learning & Scientific Computing: PyTorch, Hugging Face, Triton, Norse, NumPy/SciPy, Dask, OpenMP, Nsight profiling

Backend & Data: FastAPI, SQLAlchemy, REST APIs, gRPC/Protobuf, MCP, SQLite/FTS5, Temporal, Kafka, Spark/Databricks, Cassandra, Vue 3, pytest, unit/integration/load testing

Cloud & Infrastructure: Docker, Kubernetes, Azure, Terraform, SLURM, Linux, Git, GitHub Actions, CI/CD, Prometheus/Grafana, Splunk

Hardware/EDA: Xilinx Vivado, Cadence Virtuoso, SPICE, DRC/LVS, parasitic extraction